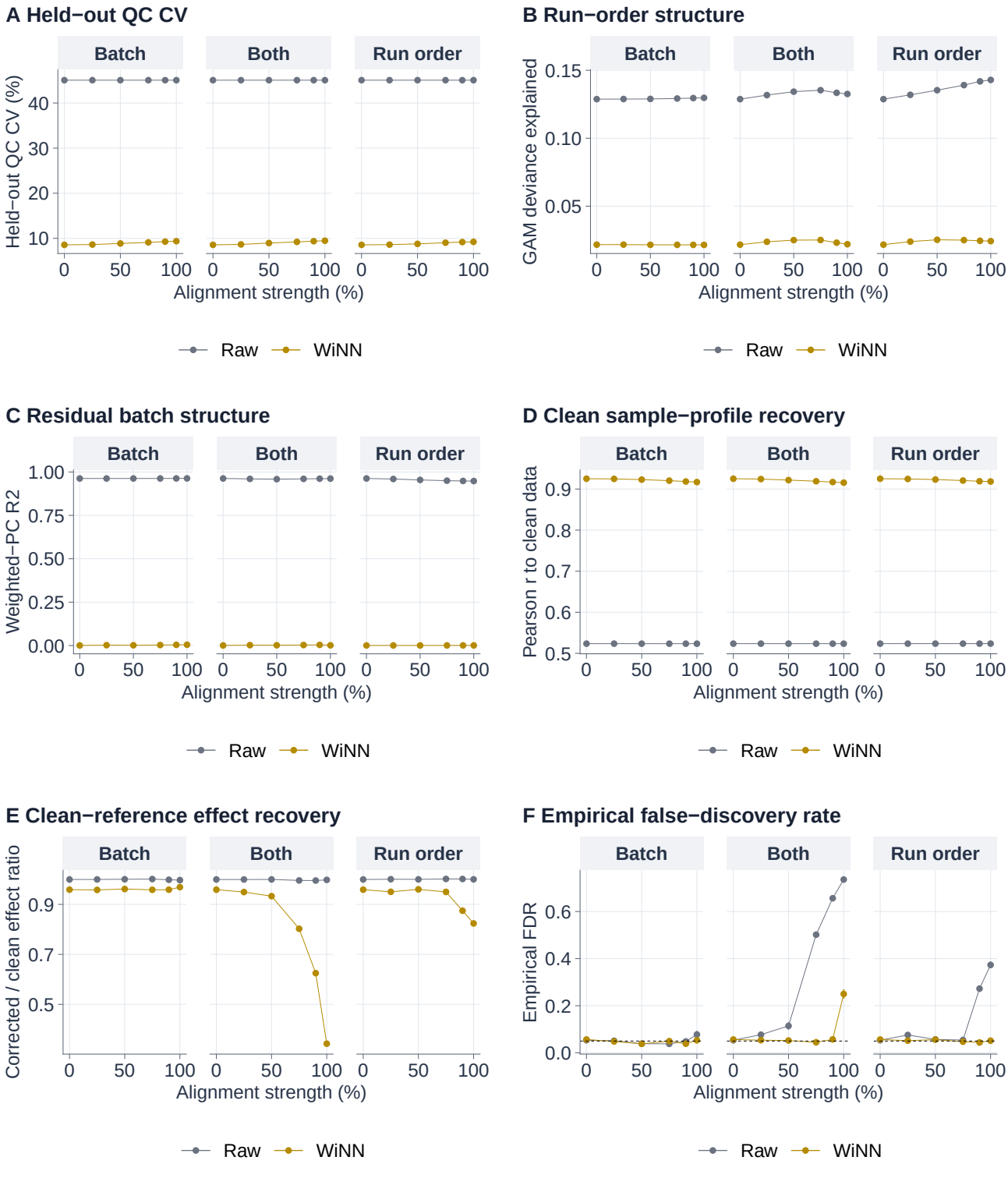

A Held−out QC CV
Batch
Both
Run order
Held−out QC CV (%)
40
30
20
10
0
50
100
Alignment strength (%)
Raw
WiNN
B Run−order structure
GAM deviance explained
0.15
0.10
0.05
C Residual batch structure
Weighted−PC R2
1.00
0.75
0.50
0.25
0.00
D Clean sample−profile recovery
Pearson r to clean data
0.9
0.8
0.7
0.6
0.5
E Clean−reference effect recovery
Corrected / clean effect ratio
F Empirical false−discovery rate
Empirical FDR
0.6
0.4
0.2
0.0